NEW CONSTRUCTIONS, OBSTRUCTIONS, AND MULTIPLIER STRUCTURE FOR ERDŐS PROBLEM 634

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ABSTRACT. Let $\mathcal{S}$ be the set of integers $N > 1$ for which some nondegenerate Euclidean triangle can be tiled by N congruent triangles. This paper continues the determination of the prime members of $\mathcal{S}$ and develops a framework for the full Erdős Problem 634. We give exact-coordinate certificates proving $88, 189 \in \mathcal{S}$, and an exact computer-assisted exhaustion proving $33 \notin \mathcal{S}$. The 189-tiling settles the first unknown multiplier on the Group 1 ray $21n^2$ and, with known constructions, shows that every multiplier $n \geq 2$ occurs on that ray.

For a primitive 120° tile with sides $c^2 = a^2 + ab + b^2$, we introduce a Laurent-polynomial boundary invariant whose quotient is $\mathbb{Z}/(3ab)$. It subsumes the ideal-membership obstructions from the known signed boundary characters and gives the exact arithmetic necessity on all six Group 2 target shapes. We also prove that every Group 2 multiplier set is a numerical semigroup with an explicit finite generator window. A trapezoid construction improves known sufficient thresholds; exact positive certificates and a one-implementation exhaustive search determine the entire one-leg family for the tile $(3, 5, 7)$. Supporting transfer and ordered-segment results isolate, but do not solve, the remaining geometric seam-propagation problem. Detailed computational qualifications, a result ledger, and a provisional programme below 200 are separated into the appendices.

1. Introduction

An integer $N > 1$ belongs to $\mathcal{S}$ if at least one nondegenerate triangle admits a dissection into N pairwise interior-disjoint triangles congruent to one another; reflections are allowed and the dissection need not be edge-to-edge. Erdős Problem 634 asks for a characterization of $\mathcal{S}$ [8, 14]. Classical constructions give

$$n^2, \qquad 2n^2, \qquad 3n^2, \qquad 6n^2, \qquad n^2 + m^2 \in \mathcal{S}$$

in their natural ranges [13, 14]. Recent classification theorems of Laczkovich, Beeson, and Zhang reduce all remaining possibilities to a small number of geometric branches [11, 12, 6, 7].

This paper is a sequel to [10]. Three contemporaneous manuscripts independently obtained the prime classification

$$p = 2, \qquad p = 3, \qquad \text{or} \qquad p \equiv 1 \pmod 4.$$

They use complementary direct and classification-based arguments [9, 10, 5, 4]; no claim of exclusive priority is intended. Beeson's four scalene 120° count formulae also confirm four rows of Table 1. These manuscripts settle only the prime subproblem; the full characterization of $\mathcal{S}$ remains open.

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The full problem is substantially harder than the prime case. Arithmetic classification places a candidate on a square ray $N = Cm^2$, but it usually does not decide which multipliers m are geometrically realizable. The three principal contributions of this paper are:

(1) exact, independently checkable certificates for the new counts 88 and 189, together with a carefully scoped computer-assisted exclusion of 33;
(2) a single Laurent quotient that closes the $\chi(j,m) = (-1)^j z^m$ ideal-membership programme for 120° tiles; and
(3) additive and finite-generation structure for the multiplier sets, turning every Group 2 ray into finitely many exact tiling questions.

Exact trapezoid, transfer, and ordered-segment results support this programme and identify the remaining geometric obstruction.

Evidence levels. A result labelled *proved* has a mathematical proof here or in a cited source. A *computer-assisted theorem* depends on an exact exhaustive search whose completeness argument and reproducibility data are given in Appendix A. A *verified certificate* is a positive tiling rechecked from exact coordinates, independently of the search history. The label *conditional* is reserved for consequences of the unfinished forced-layer lemma in Appendix F. No conditional statement is used to prove an unconditional result.

2. The classified branches and the prime case

We use the classification in the same form as [10]. If the tile has incommensurable angles and the tiled triangle is neither a reptile nor covered by the isosceles or equilateral classifications, the non-similar possibilities lie in two groups:

$$3\alpha + 2\beta = \pi \quad \text{(Group 1)}, \qquad \gamma = \frac{2\pi}{3} \quad \text{(Group 2)}.$$

Beeson–Zhang rationality then makes the relevant tile sides commensurable [7]. The Group 2 tile can consequently be scaled to a primitive integer triple

$$c^2 = a^2 + ab + b^2, \qquad \gcd(a,b,c) = 1, \tag{1}$$

where c is opposite the 120° angle. Throughout the paper, the tile angles α, β, γ are opposite the tile sides a, b, c, respectively.

Lemma 1 (Arithmetic of primitive Group 2 triples)**.** *For a primitive positive solution of* (1)*, the integers a, b, c are pairwise coprime, $3 \nmid c$, and both c and $a - b$ are units modulo $3ab$.*

Proof. If a prime divides any two of a, b, c, the norm equation makes it divide the third, contrary to primitivity. Thus the triple is pairwise coprime. If $3 \mid c$, reduction modulo 3 gives $a \equiv b \pmod 3$. If $3 \mid a$, then also $3 \mid b$, again impossible. Otherwise write $b = a + 3t$; then

$$a^2 + ab + b^2 = 3(a^2 + 3at + 3t^2) \equiv 3 \pmod 9,$$

whereas $c^2 \equiv 0 \pmod 9$, a contradiction. Hence $3 \nmid c$. Finally, pairwise coprimality gives $\gcd(a - b, ab) = 1$, and $(a - b)^2 \equiv c^2 \not\equiv 0 \pmod 3$ gives $3 \nmid (a - b)$. The stated unit claims follow. □

The six Group 2 target shapes are listed in Table 1. We use

$$U = a + 2b, \qquad V = 2a + b, \qquad W = a + b.$$

Rows I–V are the five non-equilateral shapes in Laczkovich's table, with the interchange $(\alpha, a) \leftrightarrow (\beta, b)$ allowed; row E is the equilateral target.

The prime case is now unusually well cross-checked. The proof in [10] uses scale integrality and the count formulae; [9] adds a signed boundary obstruction; and [5] derives the four scalene

row	target data	necessary scale condition	tile count
I	sides $k(c, c, U)$	$b \mid k$	m^2bU
II	sides $k(c^2, Uc, 3bW)$	none beyond $k \in \mathbb{N}$	$3m^2UW$
III	sides $k(ac, bV, Wc)$	none beyond $k \in \mathbb{N}$	m^2VW
IV	sides $k(a, c, W)$	$b \mid k$	m^2bW
V	sides $k(aU, bV, c^2)$	none beyond $k \in \mathbb{N}$	m^2UV
E	equilateral, side L	$ab \mid L$	m^2ab

TABLE 1. The Group 2 rays. In rows I and IV, $k = bm$; in row E, $L = abm$. In the other rows $m = k$. The arithmetic necessity is proved in Section 4; known constructions realize a tail, but generally not every multiplier.

Group 2 cases independently, delegating the other branches to the cited classification papers. We do not repeat that proof. Its role here is to close one slice of $\mathcal{S}$ and to fix the imported framework on which the composite investigation rests.

3. New exact tilings and the exclusion of 33

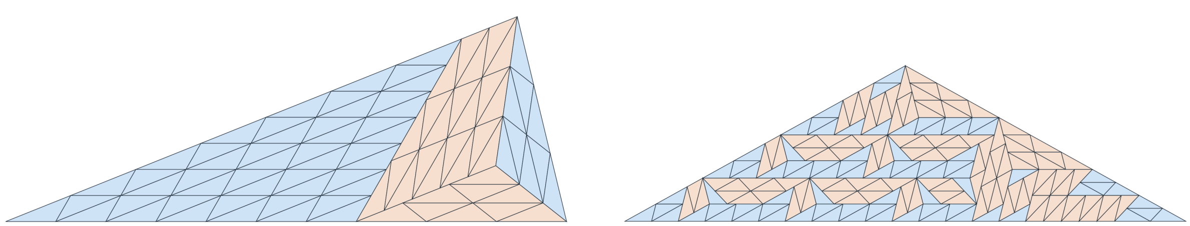
Theorem 2 (Verified exact certificates)**.** *The following tilings exist.*

(i) *A triangle with side lengths* $(21, 55, 56)$ *is tiled by* 88 *congruent copies of the triangle* $(3, 5, 7)$.

(ii) *A triangle with side lengths* $(36, 36, 63)$ *is tiled by* 189 *congruent copies of the triangle* $(2, 3, 4)$.

Consequently $88m^2, 189m^2 \in \mathcal{S}$ *for every* $m \geq 1$.

Proof. The exact vertex coordinates are the certificates identified in Appendix A.1. For each certificate, an exact checker verifies the three squared side lengths of every tile, containment in the target, pairwise interior-disjointness by rational polygon clipping, and equality of the exact total area with the target area. This proves that the listed triangles are the union of the listed congruent tiles. Subdividing every tile quadratically into m^2 similar copies gives the final assertion. □



88 copies of $(3, 5, 7)$ tile $(21, 55, 56)$. 189 copies of $(2, 3, 4)$ tile $(36, 36, 63)$.

FIGURE 1. The two new tilings. These images are renderings of the exact coordinate certificates, not geometric evidence in place of them.

The 88-tiling is row III of Table 1 at multiplier one: $VW = (2a + b)(a + b) = 11 \cdot 8 = 88$. Zhang's general construction for this row, through its published uniform threshold, first produces $7128 = 88 \cdot 9^2$ tiles for $(3, 5, 7)$ [15]. Thus the exact example reduces the smallest known count on this ray by a factor 81.

The 189-tiling is Group 1 shape 5 for $(p,q)=(1,2)$, whose ray is $21n^2$; see Section 6. The known 84- and 336-tilings give multipliers 2 and 4 [2], while the new tiling gives multiplier 3. The addition theorem below therefore yields:

Corollary 3 (Computer-assisted)**.** *On the Group 1 shape-5 ray $21n^2$, a tiling exists exactly for $n \geq 2$.*

Proof. Beeson's Table 5 reports $n=1$ as negative on the basis of a computer boundary search [2]. Our exact polygon search supplies the documented proof surface: it exhausts the 21-tile shape-5 target in 415 states. The multipliers 2 and 3 occur, and Proposition 11 makes the multiplier set additively closed. Since $\langle 2,3\rangle = \{n \geq 2\}$, every remaining multiplier occurs. □

Theorem 4 (Computer-assisted)**.** $33 \notin \mathcal{S}$.

Proof. The published classification, Corollary 8, Propositions 14 and 15, and the finite branch audit in Appendix A.2 reduce a hypothetical 33-tiling to one target: the isosceles triangle $(21,21,33)$ tiled by $(5,3,7)$; this agrees with Beeson's effective isosceles enumeration [3]. The primary exact polygon search exhausts this instance in 6479 states. A replay with the general segment-semigroup table and boundary-invariant pruning disabled also returns `EXHAUSTED`, in 22850 states; feasible-wedge and short-convex-edge tests remain. The convex-corner branching lemma and exact subtraction argument in Appendix A.2 prove that the search tree is complete. No second implementation is used in this proof. □

Remark 5. The boundary invariant of Section 4 does not exclude this candidate: it satisfies the Laurent boundary congruence and even admits the required exact-N signed inventory. Theorem 4 is therefore a small but concrete demonstration that any final solution needs interior order, incidence, or connectivity information.

4. A Laurent boundary quotient

Lemma 6 (Boundary integrality)**.** *If a convex polygon is tiled by a triangle with integer side lengths, every side of the polygon has integer length and is a concatenation of complete tile edges.*

Proof. Let ℓ support one side of the polygon. Every tile lies in the same closed half-plane bounded by ℓ. Hence a tile meeting the relative interior of that side in a nondegenerate segment meets ℓ in a face, and that face must be a complete tile edge. Distinct such edges have disjoint relative interiors, while finitely many of them cover the polygon side. Their integer lengths therefore add to its length. No edge-to-edge hypothesis is used. □

In a Group 2 tiling, every directed edge direction has a unique expression

$$\theta = j\pi/3 + m\alpha, \qquad j \in \mathbb{Z}/6, \quad m \in \mathbb{Z},$$

after an anchor is chosen; here α/π is irrational [7]. Put $d=a-b$ and define the character

$$\chi(j,m) = (-1)^j z^m \in \mathbb{Z}[z,z^{-1}]^\times.$$

For a directed segment PQ of direction θ, set $B(PQ)=|PQ|\chi(\theta)$, initially in $\mathbb{R}[z,z^{-1}]$, and extend B additively to oriented polygonal chains. By Lemma 6, complete tile and target boundaries have integral coefficients.

Theorem 7 (Laurent boundary quotient)**.** *Let a triangle be tiled by a primitive Group 2 tile satisfying* (1)*. Then*

$$B(\partial T) \in I = (d+cz,\ d+cz^{-1}) \subseteq \mathbb{Z}[z,z^{-1}], \tag{2}$$

and

$$\mathbb{Z}[z, z^{-1}]/I \cong \mathbb{Z}/(3ab), \qquad z \longmapsto -dc^{-1}. \tag{3}$$

Thus (2) *is equivalent to the single evaluation*

$$B(\partial T)\big|_{z=-dc^{-1}} \equiv 0 \pmod{3ab}. \tag{4}$$

Every ideal-membership obstruction obtained by applying a signed, root-of-unity, or other ring specialization to this χ-based invariant is a specialization of (2)*. This statement does not include the stronger requirement that a boundary be represented by exactly N allowed tile terms.*

Proof. The counterclockwise boundary of a direct tile, with its a-edge anchored, has value

$$d + cz^{-1};$$

a reflected tile has value $d + cz$, up to multiplication by a Laurent unit. Subdivide every tile edge at every tiling vertex. In this common refinement, each elementary interior segment is incident with exactly one tile on each side, so it occurs twice with opposite orientations and cancels. This proves (2); no edge-to-edge hypothesis is being imposed.

Since $c^2 - d^2 = 3ab$,

$$c(d + cz) - dz(d + cz^{-1}) = (c^2 - d^2)z = 3abz,$$

and z is a unit, so $3ab \in I$. Moreover c and d are invertible modulo $3ab$ by Lemma 1. Let ε evaluate z at $-dc^{-1}$ modulo $3ab$. Both generators of I lie in $\ker \varepsilon$, the second because $c^2 \equiv d^2 \pmod{3ab}$. Conversely, in the quotient by I, the relation $3ab = 0$ makes c a unit and $d + cz = 0$ makes every Laurent monomial a constant. Constants are reduced precisely modulo $3ab$, because $I \cap \mathbb{Z} \subseteq \ker \varepsilon \cap \mathbb{Z} = (3ab)$ and $3ab \in I$. Thus $\ker \varepsilon = I$, proving (3) and (4). Applying a ring specialization to the ideal containment proves the final, deliberately limited assertion. □

Corollary 8 (Exact Group 2 arithmetic)**.** *The six necessary conditions and count rays in Table 1 hold for every Group 2 tiling.*

Proof. Evaluate the counterclockwise target boundary in (4). Up to multiplication by units modulo $3ab$, the six residues are

row	I	II	III	IV	V	E
$B(\partial T)$	$3a^2k$	0	0	$3ak$	0	$3L$.

Since $\gcd(a, b) = 1$, the nonzero entries give $b \mid k$ in rows I and IV and $ab \mid L$ in row E. The area equations give the counts in Table 1. The primitive target side vectors make the remaining scale integral; detailed row polynomials and gcd checks are recorded in Appendix B. □

Remark 9 (What the invariant closes)*.* The quotient proves Zhang's divisibility conjecture for equilateral 120° tiles, including the nonsquarefree cases not covered by his lemma, and the same proof works for 60° tiles after replacing $c^2 = a^2 + ab + b^2$ by $c^2 = a^2 - ab + b^2$ and $d = a - b$ by $a+b$: the one-tile polynomials become, up to units, $d - cz^{-1}$ and $d - cz$, while $c^2 - d^2 = -3ab$. It also removes the congruence hypothesis from Zhang's row-V count formula [15]. Conversely, all ideal-membership obstructions obtained by specializing this Laurent character are exhausted. Exact-N coefficient bounds and non-character invariants lie outside this claim.

5. Construction thresholds and additive multiplier sets

For $x, L > 0$, an *ideal trapezoid* $T(x, L)$ has parallel sides x and $x + L$, base angles $\pi/3$, and two legs of length L.[1] Its Laurent boundary is $3L$, independent of x, so Theorem 7 gives the necessary condition $ab \mid L$.

Lemma 10 (Trapezoid construction)**.** *Let $R = (a, b, c)$ be a primitive $120°$ tile. If*

$$x, L \in \mathbb{N}, \qquad ab \mid L, \qquad x - (a^2 + b^2) \in a\mathbb{N}_0 + b\mathbb{N}_0,$$

then $T(x, L)$ is tiled by

$$\frac{L(2x + L)}{ab}$$

copies of R.

Proof. Zhang constructs the basic trapezoid $T(a^2 + b^2, ab)$ and the $\pi/3$-parallelograms of side ab and base a or b [15, Lemmas 1–2]. Write $L = kab$ and slice $T(x, L)$ into k layers of leg ab. In layer i, remove a copy of the basic trapezoid. The remainder is a parallelogram whose base is

$$x + (i - 1)ab - (a^2 + b^2) \in a\mathbb{N}_0 + b\mathbb{N}_0,$$

so it is tiled by concatenating Zhang's two parallelograms. The area ratio gives the stated number of tiles. □

For $x = rab$, the semigroup condition in Lemma 10 is equivalent to

$$r \geq R(a, b) := \left\lceil \frac{a}{b} \right\rceil + \left\lceil \frac{b}{a} \right\rceil.$$

Indeed, if $rab - a^2 - b^2 = Aa + Bb$, reduction modulo b and a gives $A = ub - a$ and $B = va - b$ for integers $u \geq \lceil a/b \rceil$, $v \geq \lceil b/a \rceil$, and then $r = u + v$. The converse follows from the same formulae. In Zhang's cyclic equilateral construction choose the three trapezoid parameters $R, R, m - 2R$; all are at least R exactly when $m \geq 3R$. This gives an equilateral tiling for every such multiplier. At $(a, b, c) = (32, 45, 67)$ this gives the smaller multiplier $m = 9$.[2] The count 116640 is already classical, so this changes the construction theory rather than $\mathcal{S}$ itself.

Proposition 11 (Multiplier addition)**.** *Let Δ be a triangle and R a tile. Suppose one vertex of Δ has adjacent sides L_1, L_2, while two adjacent sides e_1, e_2 of R meet at that angle or its supplement, and $L_1/e_1, L_2/e_2 \in \mathbb{N}$. If $r, s \in \mathbb{N}$ and $r\Delta$ and $s\Delta$ are tiled by R, then $(r + s)\Delta$ is tiled by R.*

Proof. Write $\Delta = \mathrm{conv}(O, A, B)$. In $(r + s)\Delta$, put $P = rA$, $Q = rB$, and $D = rA + sB$. The three regions

$$\triangle OPQ, \qquad P + s\Delta, \qquad \mathrm{conv}(P, Q, (r + s)B, D)$$

have disjoint interiors and fill $(r + s)\Delta$; the last is a parallelogram with side lengths $r|B - A|$ and $s|B|$. Labeling the chosen corner of Δ as B makes these rL_1 and sL_2. Subdivide the parallelogram into an $r(L_1/e_1)$ by $s(L_2/e_2)$ grid of parallelograms with sides e_1, e_2, and split every cell along the diagonal determined by whether the included angle is the tile angle or its supplement. The two halves are congruent copies of R by SAS. No edge-to-edge hypothesis is used. □

[1] Zhang [15] interchanges the leg and long-side symbols; here L is the leg.

[2] The threshold printed in [15, Conjecture 2] is 12, so that conjectured threshold is not sharp.

For a fixed ray, let $\mathcal{A}$ be the set of positive multipliers that admit a tiling. Here a *numerical semigroup* means a cofinite additive submonoid of $\mathbb{N}_0$. The hypotheses of Proposition 11 hold on all six Group 2 rows and on Group 1 shapes 4 and 5; the integer ratios are given in Appendix C.

Corollary 12 (Finite generator windows)**.** *For every primitive Group 2 tile and every row of Table 1, $\mathcal{A} \cup \{0\}$ is a numerical semigroup. More precisely:*

(i) *in rows E, I, III, and IV, every $m \geq 3R(a, b)$ occurs and*

$$\mathcal{A} \cup \{0\} = \big\langle \mathcal{A} \cap [1, 6R(a, b) - 1] \big\rangle;$$

(ii) *in rows V and II, every $s \geq \lceil 3R(a, b)/2 \rceil$ occurs and*

$$\mathcal{A} \cup \{0\} = \big\langle \mathcal{A} \cap [1, 2\lceil 3R(a, b)/2 \rceil - 1] \big\rangle.$$

Thus each Group 2 ray is determined by finitely many exact tiling questions.

Proof. Additive closure is Proposition 11. The equilateral tail $m \geq 3R(a, b)$ was proved immediately above from Lemma 10 and Zhang's cyclic construction. Zhang's Propositions 8–10 transfer that tail to rows I, III, and IV [15, Propositions 8–10]. For rows V and II, a bisector dissection transfers an equilateral tiling of side $2sab$ to row V at multiplier s; the dissection and count identity are in Appendix C. Zhang's Proposition 11 transfers the same interval to row II [15, Proposition 11]. Finally, if an additively closed set contains every integer at least C, then each $n \geq 2C$ is $C + (n - C)$ and induction shows that the set is generated by its members below $2C$. The complement is finite, so $\mathcal{A} \cup \{0\}$ is a numerical semigroup. □

Theorem 13 (Computer-assisted trapezoid classification)**.** *For the tile* $(3, 5, 7)$*,* $L = 15$*, and every integer* $x \geq 1$*,*

$$T(x, 15) \text{ is tileable} \iff x \in \{29, 31, 32\} \cup \{x \in \mathbb{N} : x \geq 34\}.$$

Proof. The boundary inventory first forces $x \in \langle 3, 5, 7 \rangle$. Lemma 10 covers $x = 34, 37, 39, 40$ and every $x \geq 42$; explicit certificates cover

$$x = 29, 31, 32, 35, 36, 38, 41.$$

The boundary inventory excludes positive $x \notin \langle 3, 5, 7 \rangle$. The exact searcher exhausts precisely the remaining 27 cases

$$\langle 3, 5, 7 \rangle \cap [1, 33] \setminus \{29, 31, 32\}.$$

The campaign comprises 34 searches and 5,699,235 states, with no skipped placement warnings. Appendix A.5 gives the case table and the completeness scope. In particular, the negative half presently depends on one exhaustion implementation; the seven positive cases are independently checkable coordinate certificates. □

Theorem 13 gives the complete local picture for this one-leg family.[3] The necessary divisibility $ab \mid L$ remains valid.

[3]The literal inequality in [15, Conjecture 5] does not hold as printed and already conflicts with the basic construction in [15, Lemma 1].

6. The remaining arithmetic branches

For Group 1, normalize $a/c = p/q$ with $0 < p < q$ and $\gcd(p, q) = 1$. A primitive tile has sides

$$(pq,\ q^2 - p^2,\ q^2).$$

The five target shapes lie on the rays in Table 2. Shapes 1–3 are characterized by Beeson [2]. Shapes 4–5 are necessity statements; their sufficiency remains open in general.

Proposition 14. *For every Group 1 shape-4 tiling, the natural multiplier m satisfies $q \mid m$; hence its count has the shape-4 form in Table 2. Every shape-5 tiling has the shape-5 count in that table, without an additional parity factor. The multiplier sets for shapes 4 and 5, with zero adjoined, are additive submonoids of $\mathbb{N}_0$.*

Proof. For shape 4, an exact target insertion gives equal side $X = (q^2 - p^2)qm$. Attach a qm-scaled tile along that side. The union is a shape-1 target with count $(2q^2 - p^2)m^2$. Beeson's necessary-and-sufficient shape-1 theorem says its count is $q^2(2q^2 - p^2)n^2$, forcing $m = qn$ [2, Theorem 7]. For shape 5, cancellation in the Group 1 boundary ideal forces the target scale C to be divisible by $q^2 - p^2$, giving the listed count. The factorization and target insertion are recorded in Appendix B. Additive closure is Proposition 11. $\square$

Unlike Corollary 12, this proposition supplies neither a tail nor an a priori finite generator window for shapes 4 and 5.

shape	target angles	necessary count ray
1	$(2\alpha, \beta, \alpha + \beta)$	$q^2(2q^2 - p^2)n^2$
2	$(2\alpha, \alpha, 2\beta)$	$(2q^2 - p^2)(3q^2 - p^2)n^2$
3	$(\beta, \beta, 3\alpha)$	$q^2(3q^2 - p^2)n^2$
4	$(\alpha + \beta, \alpha + \beta, \alpha)$	$q^2(q^2 - p^2)n^2$
5	$(\alpha, \alpha, \alpha + 2\beta)$	$(q^2 - p^2)(2q^2 - p^2)n^2$

Table 2. Group 1 rays. In shape 4 the natural scale m is divisible by q, so $m = qn$; in shape 5 there is no additional parity factor, including when p and q are both odd.

We do not use a further divisibility clause in Beeson's Theorem 19.[4]

The remaining isosceles branch has tile angle $\gamma = 2\alpha$ [3]. With coprime integers $u < v < 2u$, its primitive side triple is

$$(a, b, c) = (u^2,\ v^2 - u^2,\ uv).$$

Proposition 15 ($\gamma = 2\alpha$ ray)**.** *Every tiling of a non-similar isosceles target in this branch has*

$$N = (v^2 - u^2)r^2$$

for an integer $r \geq 2$.

Proof. The exact boundary ideal forces the equal side of the isosceles target to have the required integral scale; inserting it into the area equation gives the stated square ray. At $r = 1$, the target boundary is too short to carry the necessary tile edges, giving $r \geq 2$. The full boundary insertion and capacity argument are given in Appendix B. $\square$

[4]The printed condition $g \mid M$ in arXiv:1206.2229v3 [2] is violated by 12 of the 15 rows in that paper's Table 5 and by the exact 189-tiling of Theorem 2. The ray formula in Proposition 14 does not depend on this clause.

Remark 16. A stronger assertion $r \geq 3$ is part of the conditional forced-layer program, not Proposition 15. This distinction affects several small values and is made explicit in Appendix F.

7. Full-side transfers and ordered segment data

A *one-patch row transfer* adjoins a quadratically tiled t-scaled copy of the tile along one complete target side, where $t \in \mathbb{N}$, and straightens one seam endpoint. The following result closes this particular construction mechanism; it does not classify partial-side or multi-patch attachments.

Write $X(a, b; m)$ for the Group 2 target in row $X \in \{E, I, II, III, IV, V\}$ of Table 1, with normalized multiplier m, and write $S_i(n)$ for the Group 1 target in shape i of Table 2, with its displayed normalized multiplier n.

Proposition 17 (One-patch row-transfer calculus)**.** *Under the standing incommensurable-angle hypotheses, the complete row-to-row lists, up to reflection and $(a, \alpha) \leftrightarrow (b, \beta)$, are as follows:*

$$\begin{aligned}\text{Group 2:}\quad & E(a,b;m) \longrightarrow IV(a,b;m),\\ & IV(a,b;m) \longrightarrow I(a,b;m),\ III(b,a;m),\\ & V(a,b;m) \longrightarrow II(a,b;m),\\ & IV(a,b;as) \longrightarrow IV(a,b;cs),\\ & V(a,b;as) \longrightarrow III(a,b;cs),\\ & V(a,b;bs) \longrightarrow III(b,a;cs);\end{aligned}$$

and

$$\begin{aligned}\text{Group 1:}\quad & S_4(n) \longrightarrow S_1(n),\ S_5(n),\\ & S_1(n) \longrightarrow S_2(n),\ S_3(n),\\ & S_1(bs) \longrightarrow S_1(cs),\\ & S_4(ps) \longrightarrow S_4(qs).\end{aligned}$$

No Group 2 row-to-row transfer of this kind lands in row V, and no Group 1 transfer of this kind starts from S_5.

Proof. The finite angle enumeration, attachment scales, merged sides, and count identities are given in Appendix C. That enumeration also contains the uninformative cases in which the union is itself a scaled copy of the tile; these are reptilings, not transfers among the named non-similar target rows. □

The transfer graph explains two persistent asymmetries. A row-V target has no incoming one-patch edge, while shape 5 has no outgoing one-patch edge. Consequently those two branches require a multi-patch, partial-side, or interior-seam construction.

Proposition 18 (Relation lattice and pure-edge words)**.** *Let (a, b, c) be a primitive positive solution of $c^2 = a^2 + ab + b^2$. After possibly interchanging a and b, there are coprime integers $r > s > 0$, with $r \not\equiv s \pmod 3$, such that*

$$a = r^2 - s^2, \qquad b = 2rs + s^2, \qquad c = r^2 + rs + s^2.$$

Put

$$R_c = (r + s, s, -r), \qquad R_b = (s, -r, s).$$

Then:

(i) R_c, R_b *form a* $\mathbb{Z}$*-basis of*

$$\{(x, y, z) \in \mathbb{Z}^3 : ax + by + cz = 0\}.$$

Thus every signed equality of unordered full-edge inventories along a collinear segment has two unique integer charges.

(ii) *Orient a chain of consecutive* a*-edges from left to right and suppose all supported tiles lie in the same half-plane. With* $\rho = e^{i\pi/3}$*, call the third vertex over* $[x, x + a]$

$$A : x + a + b\rho, \qquad B : x + b\rho^2.$$

Every support word has the form $B^j A^{n-j}$ *and hence at most one defect. The analogous assertion holds for a chain of* b*-edges.*

(iii) *At the possible* $B \to A$ *defect, the two exposed third vertices and the shared endpoint bound a wedge with sides*

$$(c, c, 2a + b).$$

Its area divided by the tile area is $(2a + b)/a$*, which is never an integer.*

Proof. See Appendix D. □

The proposition is deliberately one-dimensional. The two exposed c-edges at a defect are seeds, not yet proved paths, and the nonintegral wedge is not a detachable row-I macrotile. A complete seam theorem would still have to prove continuation through every encountered vertex, embeddedness and noncrossing of suitable outermost paths, pairing or boundary termination, geometric realization of the relation charges by strips, successive exposure of peelable layers, and termination with finitely describable interfaces. A finite inventory basis alone does not imply any of these statements.

Remark 19 (Herdt's 720-tiling). The orientation-changing parallelogram construction exhibited by Beeson [3, Fig. 22] is made explicit in Appendix C. A vector-figure audit of Herdt's $(4, 5, 6)$ construction in [3, Fig. 24] finds ten quadratic patches and three orientation-changing blocks, each split into two uniform subparallelograms; their count inventory sums to 720. This is a check of the displayed macro inventory, not an exact coordinate certificate for the figure.

8. Consequences and the remaining geometric gap

The principal conceptual consequence is negative but useful: after Theorem 7, further specializations of the same $\chi(j, m) = (-1)^j z^m$ ideal-membership invariant will not settle the surviving multipliers. Promising next objects must retain ordering or incidence, or impose exact-term constraints absent from ideal membership. Three such structures are already available:

(1) orient every c-edge of a Group 2 tile from its α endpoint to its β endpoint. The resulting flow is conserved at every non-corner vertex and forces directed c-edge paths between charged target corners; such a path may run along the target boundary;
(2) row III at multiplier one is exactly a two-grid staircase problem: half of a $(2a + b) \times (a + b)$ parallelogram is cut along a legal c-edge diagonal. The 88-tiling is the first worked instance.
(3) every unordered full-edge segment mismatch has the two unique charges of Proposition 18, while a pure a- or b-edge support word has at most one ordered defect.

These facts do not yet exclude or construct a new multiplier, but they name finite, ordered geometric objects that a final argument can classify. Their short proofs are included in Appendix D.

9. Conclusion

The results above change the shape of the full problem in four ways. First, the small exact tilings at 88 and 189 show that published construction thresholds can be far from sharp. Second, the Laurent quotient settles the χ-based ideal-membership arithmetic, sharply identifying what that boundary method can and cannot decide. Third, multiplier addition turns every Group 2 ray into a numerical semigroup determined by an explicit finite window. Fourth, the complete one-patch transfer graph and the relation/word normal forms isolate the next genuinely geometric object: an ordered seam whose continuation, pairing, and interfaces must be controlled in the presence of T-junctions. Thus the Group 2 branch reduces to explicit finite per-ray windows; the other branches still require comparable construction or obstruction theorems before such a finite reduction is available for the full Erdős Problem 634.

The immediate mathematical targets are small and concrete: decide the first unknown multipliers on the row-I, row-IV, row-V, Group 1 shape-5, and $\gamma = 2\alpha$ rays; close or replace the forced-layer induction; prove a geometrically realizable seam calculus rather than only its inventory charges; and export search-independent refutation certificates for negative computations. The manuscript is structured so that those results can be added without changing the distinction between proved and conditional claims.

Appendix A. Exact certificates and positive verification

Data and code availability. The accompanying arXiv-ready source package contains an `anc` directory with the verification manifest, exact certificate data, the primary $N = 33$ search and two stored exhaustion outputs, the corrected separately written diagnostic searcher and its regression witness, and the arithmetic audit programs and captured outputs. Computational paths displayed below are relative to the root of that ancillary tree. The manifest records exact commands, environments, qualifications, and SHA-256 digests.

A.1. **Certificate semantics.** Coordinates are pairs (x, w) representing $(x, w\sqrt{D})$, with rational x, w. Here $D = 3$ for Group 2 and $D = 15$ for the displayed Group 1 $(2, 3, 4)$ instance. The verifier reads only the tile triangles from a JSON certificate; it does not read the search tree. It reconstructs their convex hull and, when supplied with the theorem's expected data, asserts the number of hull corners and the exact multiset of target side lengths. For the seven trapezoid certificates it further checks the cyclic order of the four side lengths, parallelism of the two unequal bases, and both diagonal lengths; thus the asserted hull is the intended ideal trapezoid $T(x, 15)$, not merely a quadrilateral with the same side multiset. It checks:

(1) each tile has the required three squared side lengths;
(2) every tile is contained in the reconstructed convex target;
(3) every pair of tiles has intersection of exact area zero; and
(4) the sum of the exact tile areas equals the exact target area.

certificate	tiles	chirality split	pairs checked	SHA–256
$N = 88$	88	58/30	3828	`163327fc06cf8f98…aefbbd`
$N = 189$	189	77/112	17766	`28f33084443ba303…d5096`

Table 3. Exact-certificate audit.

The full SHA–256 digests are

$N = 88$: `163327fc06cf8f98639b38743e8d796c9fdcd86d722485229a8f047e94aefbbd`
$N = 189$: `28f33084443ba303e6758511e616c3f9bbf3de1657e3422eee6680118a0d5096`.

The files are:

```
computation/tiling_search/results/N88_tiling.json
computation/tiling_search/results/N189_tiling.json
computation/tiling_search/verify_tiling.py.
```

The SVG and plain-text coordinate exports beside the JSON files are convenience views of the same data.

A.2. **The exact exhaustion method.** The primary search selects a deterministic convex corner P of each remaining polygonal component.

Lemma 20 (Convex-corner branching)**.** *Let a polygonal region be tiled by finitely many nondegenerate triangles, and let P be a convex vertex of one remaining component. Fix either incident boundary ray at P. The tile adjacent to that ray has a vertex at P and a complete tile edge beginning at P along the ray. Therefore branching over all tile angles, both assignments of the adjacent side lengths, and both chiralities is exhaustive.*

Proof. Intersect a sufficiently small disk about P with the component. It is a wedge of angle less than π, partitioned by the sectors of the incident tiles. The point P cannot lie in a tile interior, and it cannot lie in the relative interior of a tile edge, since the wedge contains no complete line through P. Thus every incident tile has a vertex at P. The tile sector adjacent to the chosen boundary ray has one of its sides on that ray, and that side is a complete edge of the tile. The edge may continue beyond a later reflex boundary vertex; exact containment, not the length of the first boundary segment, decides whether the placement is legal. $\square$

Lemma 21 (Exact boundary subtraction)**.** *Let Δ be a triangle contained in a simple polygonal component R, with no proper crossing of their boundaries, and suppose that $\partial R \cap \partial\Delta$ contains a nondegenerate segment. This is the situation for every flush placement of Lemma 20. Split ∂R and $\partial\Delta$ at every vertex of either boundary lying on the other, form the oriented chain $\partial R - \partial\Delta$, and cancel opposite elementary edges. Tracing the remaining directed edges by taking, at each vertex, the first edge clockwise from the reverse incoming direction gives exactly the counterclockwise boundaries of the positive-area components of $\overline{R \setminus \Delta}$, including when components meet at pinch vertices.*

Proof. The split edges form a finite planar arrangement. An elementary edge cancels exactly when the two boundaries coincide there; every uncancelled edge separates $\overline{R \setminus \Delta}$ on its left from its complement on its right. Hence the remaining chain is balanced at every vertex. Because the removed triangle meets ∂R in a segment, it creates no enclosed boundary component; by the Jordan curve theorem, every positive-area component of the complement is a polygonal disk. The clockwise successor rule follows the boundary of the same local face and separates different faces at a pinch vertex. It therefore uses every directed edge once in closed counterclockwise walks, one for each component. Removing degenerate zero-area walks leaves precisely the asserted components. $\square$

The program enumerates precisely these flush placements. Subtraction is performed as an exact oriented 1-chain: all component and tile edges are split at all vertices, opposite duplicates cancel, and the remaining faces are traced by a deterministic turning rule. This handles a flush tile edge spanning several collinear boundary segments. All orientation, containment, area, and intersection predicates are rational in $\mathbb{Q}(\sqrt{D})$; the implementation is exactly Lemma 21.

Lemma 22 (Convex-end segment pruning)**.** *If a polygonal region tiled by triangles with side lengths a, b, c has a boundary segment whose two endpoints are convex, then its length belongs to $a\mathbb{N}_0 + b\mathbb{N}_0 + c\mathbb{N}_0$.*

Proof. A tile meeting the relative interior of the segment in a nondegenerate interval has a complete edge on the same supporting line. Such a tile edge cannot extend past either endpoint: extension beyond a convex endpoint leaves the local interior wedge. These complete tile edges partition the segment, so their lengths add to its length. □

A placement is pruned only by necessary conditions: integer area budget, feasible convex wedges, Lemma 22, and the proved boundary invariants. Each recursive placement removes one tile, so the tree is finite. Lemma 20 shows that exhausting every branch proves nonexistence. The implementation passes its exact geometry self-tests. For $N = 33$ it visits 6479 states with all pruning enabled and 22850 states when the general segment-semigroup table and boundary-invariant pruning are disabled. The latter run retains the feasible-wedge test and the special case of Lemma 22 that rejects a selected convex-ended edge shorter than every tile side.

A separately written diagnostic searcher is included for transparency, but its historical output is not part of the proof. A previous version rejected a flush tile edge whenever its length exceeded that of the first boundary segment on the ray. Lemma 20 shows why this restriction is invalid when the next boundary vertex is reflex. The supplied regression reaches a legal two-placement state in which the first segment has length 5 and a contained tile edge has length 7. The cap has been removed; the corrected diagnostic search has not been exhausted, so no independent-exhaustion claim is made.

A.3. **The complete $N = 33$ reduction.** For clarity, the reduction to the searched target is as follows. The classical, reptile, commensurable-angle, and right-triangle count forms do not contain 33. The exact Group 1 ray formulae leave only shape-4 formal candidates with $(p, q) = (4, 7)$ or $(16, 17)$ at natural multiplier $m = 1$; Proposition 14 excludes both because $q \nmid m$. The $\gamma = 2\alpha$ formula likewise leaves only $(u, v) = (4, 7)$ or $(16, 17)$ at $r = 1$, excluded by Proposition 15. The primitive $60°$ norm equation has no equilateral candidate at 33. Finally, a finite divisor enumeration of the six Group 2 rows leaves exactly row I for $(a, b, c) = (5, 3, 7)$ at $k = 3$, namely the target with sides $(21, 21, 33)$. This agrees with Beeson's effective isosceles enumeration [3, Theorem 12.10 and Example 2].

These branch checks are executable in `computation/verify_theory_report3_claims.py`. The primary exact-search commands, source and output hashes, interpreter version, and stored outputs are recorded in `paper/verification_manifest.md`; the same manifest identifies the diagnostic regression and its scope.

A.4. **Verification runs.** The following checks were rerun for this manuscript:

check	result
Laurent and signed-character invariant checks	all passed
threshold and auxiliary-ray arithmetic	23/23
branch and fixed-N audit	76/76
boundary-polynomial and transfer audit	37/37
Group 1 and macro arithmetic audit	18/18
construction and layer arithmetic audit	53/53
addition and conditional-layer arithmetic	27/27
one-patch and lattice bounded regressions	29/29
searcher geometry self-tests	all passed
$N = 33$ fresh primary exhaustion	6479 states, no warnings
$N = 33$ reduced-pruning replay	22850 states, no warnings
long-edge/reflex-vertex regression	passed
nine positive coordinate certificates	all passed

The exact commands and hashes are collected in `paper/verification_manifest.md`. Passing a script verifies only the claims named in that script. In particular, `verify_theory_report6_claims.py` verifies arithmetic consequences of the proposed forced-layer induction, not the missing geometric induction itself. The final audit script performs finite regression checks of the one-patch enumeration, transfer identities, relation lattice, pure-edge word, and defect-wedge formula. Its ranges include $r \leq 35$ for the Group 2 parametrization and defect tests, $p, q < 60$ for Group 1 identities, and support words of length at most 10; the complete bounds are recorded in the script and manifest. The universal statements are proved in the text, not by extrapolation from these tests. The script also checks that the manually transcribed inventory of Herdt's figure sums to 720, and records an exact mirror-placement witness to the unresolved forced-layer step. It does not certify that figure's coverage, forced-layer induction, seam propagation, macro-normalization, or semilinearity.

A.5. **The trapezoid campaign.** For $(a, b, c) = (3, 5, 7)$ and $L = 15$, the seven certificate values are:

x	29	31	32	35	36	38	41
$N = 2x + 15$	73	77	79	85	87	91	97.

Each certificate passes exact congruence, containment, all pairwise intersection, and exact-area checks. The search exhausts the remaining arithmetically possible values in the finite window: all possible $x \leq 28$, and $x = 30, 33$, together with the other cases not supplied by Lemma 10. The aggregate is 27 exhausted cases and seven tilings, 5,699,235 states, and zero warnings.

The negative result is computer-assisted and depends on one exhaustion implementation. The search algorithm has the same completeness argument as the primary polygon search and extensive local tests; the positive cases are independently checkable exact coordinate certificates. Exporting a semantic refutation DAG whose checker recomputes the complete candidate set at every state remains a desirable independent audit of the negative cases.

Appendix B. Row algebra and the other branches

B.1. **Group 2 target evaluations.** Let

$$\varepsilon : \mathbb{Z}[z, z^{-1}] \longrightarrow \mathbb{Z}/(3ab), \qquad z \longmapsto -dc^{-1}.$$

Counterclockwise traversal gives the following target polynomials and evaluations:

$$\begin{aligned}
\mathrm{I}: &\quad B = k(U - cz - cz^{-1}), && d\,\varepsilon(B) \equiv 3a^2k;\\
\mathrm{II}: &\quad B = k(3bW - c^2z^{-2} - Ucz), && d^2\varepsilon(B) \equiv 6ab\,kd^2 \equiv 0;\\
\mathrm{III}: &\quad B = k(Wc - acz^2 - bVz), && c\,\varepsilon(B) \equiv 3ab\,dk \equiv 0;\\
\mathrm{IV}: &\quad B = k((2a + b) - cz), && \varepsilon(B) \equiv 3ak;\\
\mathrm{V}: &\quad B = k(c^2 - (aU + bV)z^2), && c^2\varepsilon(B) \equiv -6ab\,kd^2 \equiv 0;\\
\mathrm{E}: &\quad B = \pm 3Lz^{m_0}, && \varepsilon(B) = \pm 3L\,\varepsilon(z)^{m_0}.
\end{aligned}$$

All congruences are modulo $3ab$. The factors c, d, and $\varepsilon(z)$ used to clear denominators are units there. This proves precisely the conditions in Table 1.

Lemma 23 (Primitivity of the Group 2 target vectors). *Each of the five integer side vectors in rows I–V of Table 1 is primitive.*

Proof. Suppose throughout that a prime ℓ divides every component of the vector in question. For row I, $\ell \mid c, U$ gives $c^2 \equiv 3b^2 \pmod{\ell}$; if $\ell = 3$ this contradicts $3 \nmid c$, and otherwise it forces

$\ell \mid a, b$. For row II, $\ell \mid c$; pairwise coprimality and $3 \nmid c$ give $\ell \nmid 3b$, so $\ell \mid W$, whence $c^2 \equiv b^2 \pmod{\ell}$, a contradiction.

For row III the vector is (ac, bV, Wc). If $\ell \mid a$, then pairwise coprimality gives $\ell \nmid b$, so $\ell \mid bV$ forces $\ell \mid V = 2a + b$ and hence $\ell \mid b$, a contradiction. Thus $\ell \nmid a$, and $\ell \mid ac$ gives $\ell \mid c$. Again $\ell \nmid b$, so $\ell \mid bV$ gives $\ell \mid V$, hence $b \equiv -2a \pmod{\ell}$. The norm equation then yields $0 \equiv c^2 \equiv 3a^2 \pmod{\ell}$; this forces $\ell = 3$, contrary to $3 \nmid c$. Row IV is immediate from $\gcd(a, c) = 1$. Finally, in row V one has $\ell \mid c$ and hence $\ell \nmid a, b$; therefore $\ell \mid U, V$. The identities $2U - V = 3b$ and $2V - U = 3a$ force $\ell = 3$, again contradicting $3 \nmid c$. □

To justify the integral scale used in Table 1, write the classified target side vector as $\lambda\mathbf{v}$, where $\mathbf{v}$ is one of the primitive integer vectors above. Lemma 6 makes every component of $\lambda\mathbf{v}$ integral. Bézout coefficients for the components of $\mathbf{v}$ then express λ as an integer combination of those integral side lengths, so $\lambda \in \mathbb{Z}$.

B.2. **Group 1 boundary factorization.** Put

$$b = q^2 - p^2, \qquad h = 2q^2 - p^2, \qquad Q = qz^2 + pz + q, \qquad \overline{Q} = qz^2 - pz + q.$$

For the primitive tile (pq, b, q^2), the two one-tile boundary polynomials factor as

$$F(z) = -pqz^3 + bz^2 + q^2 = (q - pz)Q,$$
$$G(z) = q^2z^3 + bz - pq = (qz - p)Q.$$

The common quadratic is the geometric direction relation. Set $J = (q - pz, qz - p)$. The homomorphism

$$\mathbb{Z}[z, z^{-1}] \longrightarrow \mathbb{Z}/b, \qquad z \longmapsto qp^{-1},$$

is well defined because p and q are units modulo b, and it kills both generators of J. Conversely,

$$q(q - pz) + p(qz - p) = b$$

and $q - pz = 0$ determines every Laurent monomial after reduction modulo b. Hence

$$\mathbb{Z}[z, z^{-1}]/J \cong \mathbb{Z}/b, \qquad J \cap \mathbb{Z} = (b).$$

For shape 4, boundary integrality and $\gcd(p, q) = 1$ make the target sides (Cq, Cq, Cp) with $C \in \mathbb{Z}$. Direct insertion gives

$$zB(\partial T) = CQ.$$

Every tile contribution lies in QJ. Cancelling Q in the Laurent domain gives $C \in J$, and hence $C \in J \cap \mathbb{Z} = (b)$. Write $C = bm$. The equal target side is therefore

$$X = Cq = bqm.$$

Attaching a tile scaled by qm along that side produces the shape-1 target used in Proposition 14; Beeson's shape-1 theorem then forces $q \mid m$.

For shape 5, boundary integrality and $\gcd(q^2, h) = 1$ make the target sides (Cq^2, Cq^2, Ch) with $C \in \mathbb{Z}$. Direct insertion gives

$$z^2B(\partial T) = C(q^2z^4 + hz^2 + q^2) = CQ\overline{Q}.$$

Every tile contribution lies in QJ. Cancelling Q in the Laurent domain therefore gives $C\overline{Q} \in J$. In the quotient,

$$\overline{Q}(qp^{-1}) = q^3p^{-2},$$

which is a unit, so $b \mid C$. Writing $C = bn$ in the area identity $N = C^2h/b$ gives $N = bhn^2$, with no additional parity factor even when p, q are both odd.

B.3. **The $\gamma = 2\alpha$ insertion.** Put $b = v^2 - u^2$. The two one-tile polynomials factor as

$$u^2 z^3 - bz + uv = (uz + v)(uz^2 - vz + u),$$
$$uvz^3 - bz^2 + u^2 = (u + vz)(uz^2 - vz + u).$$

Thus the residual ideal is $J = (uz + v, u + vz)$, and

$$v(uz + v) - u(u + vz) = b, \qquad \mathbb{Z}[z, z^{-1}]/J \cong \mathbb{Z}/b, \quad z \mapsto -vu^{-1}.$$

The target is isosceles with equal side X and base Y. The law of sines gives $X = ut$, $Y = vt$. Its boundary is, up to a Laurent unit, $t(uz^2 - vz + u)$. Cancelling the common quadratic from the target and tile boundaries forces $t \in J \cap \mathbb{Z} = (b)$, say $t = br$. Beeson's area equation $X^2 = Nu^2 b$ then gives $N = br^2$.

It remains to exclude $r = 1$. Then an equal target side has length ub and decomposes into complete tile edges as

$$ub = Au^2 + Bb + Cuv, \qquad A, B, C \in \mathbb{N}_0.$$

Reduction modulo u gives $u \mid B$, since $\gcd(u, b) = 1$, while the size bound gives $B \leq u$. Hence $B = 0$ or $B = u$.

If α/π were rational, then $2\cos\alpha = v/u$ would be a rational algebraic integer and hence an integer. But $1 < v/u < 2$, a contradiction. Thus $\alpha/\pi \notin \mathbb{Q}$. Since $3\alpha + \beta = \pi$ and $\gamma = 2\alpha$, the only angle multisets at an internal straight-boundary vertex are

$$\{3\alpha, \beta\}, \qquad \{\alpha, \beta, \gamma\}.$$

If $B = u$, equality forces $A = C = 0$. The side then consists of u edges of length b, each with one γ endpoint. Neither target endpoint carries γ, while each of the $u - 1$ internal vertices carries at most one: a contradiction.

If $B = 0$, let $E = A + C$. Every a- or c-edge has one β endpoint. At the apex the angle $\alpha + \beta$ is decomposed as one α and one β; choose the equal side whose incident apex tile contributes the α. Its other endpoint has angle α, so neither endpoint of this side supplies a β. Its E beta endpoints would have to occupy only $E - 1$ internal vertices, each of which carries exactly one β. This is again impossible. Thus $r \geq 2$. The argument does not imply $r \geq 3$.

Appendix C. Addition data and constructive transfers

For Proposition 11, the following corner data work at multiplier one:

ray	target adjacent sides	tile adjacent sides	integer ratios
I	bc, bU	c, b	(b, U)
II	$Uc, 3bW$	c, b	$(U, 3W)$
III	bV, cW	b, c	(V, W)
IV	bc, bW	c, b	(b, W)
V	aU, bV	a, b	(U, V), supplement
E	ab, ab	a, b	(b, a), supplement
G1 shape 4	bc, bc	b, c	(c, b)
G1 shape 5	$bc, b(2q^2 - p^2)$	c, b	$(b, 2q^2 - p^2)$

For the row-V interval transfer, bisect the two doubled base angles of the row-V target at multiplier s. Label its vertices

$$\angle A = 2\beta, \qquad \angle B = 2\alpha, \qquad \angle C = \alpha + \beta = \pi/3,$$

and let the bisectors from A and B meet at I. The sine rule in $\triangle AIB$, which has angles $\beta, \alpha, 2\pi/3$, gives $AI = acs$ and $BI = bcs$. Thus AIB is a cs-scaled copy of the tile.

Choose $P \in AC$ so that AIP is an as-scaled tile with AI as its c-side, and choose $Q \in BC$ analogously so that BIQ is a bs-scaled tile. The angles at I are respectively $\alpha, 2\pi/3, \beta$, so P, I, Q are collinear. Moreover $PI = IQ = abs$, hence $PQ = 2abs$. The angles at P and Q supplementary to the two $2\pi/3$ tile angles are both $\pi/3$, as is $\angle C$; therefore PCQ is equilateral of side $2abs$.

Quadratically tiling the three corner triangles and tiling the equilateral core gives

$$(c^2 + a^2 + b^2 + 4ab)s^2 = UVs^2,$$

where the equality uses $c^2 = a^2+ab+b^2$. Zhang's Proposition 11 then gives the same multiplier interval for row II [15].

C.1. **The complete one-patch tables.** We use ordered notation

$$\langle \theta_0, \theta_1, \theta_2 \mid x_0, x_1, x_2 \rangle,$$

where x_i is opposite θ_i. The elementary gluing rule used below is

$$\begin{aligned}\langle \theta_0, \theta_1, \theta_2 \mid x_0, x_1, x_2 \rangle &\cup \langle \phi_0, \phi_1, \phi_2 \mid ty_0, ty_1, ty_2 \rangle \\ &= \langle \theta_0, \theta_2 + \phi_2, \phi_0 \mid ty_1,\ x_2 + ty_2,\ x_1 \rangle,\end{aligned}$$

provided $x_0 = ty_0$, the corresponding seam endpoints satisfy $\theta_1 + \phi_1 = \pi$, and $\theta_2 + \phi_2 < \pi$. Thus one endpoint disappears, while the two collinear outer sides at that endpoint merge by addition.

The ordered Group 2 data are

row	angles	opposite sides
$I(m)$	$(\alpha, \alpha, \alpha + 3\beta)$	$m(bc, bc, bU)$
$II(m)$	$(\alpha, 2\alpha, 3\beta)$	$m(c^2, Uc, 3bW)$
$III(m)$	$(\alpha, 2\beta, 2\alpha + \beta)$	$m(ac, bV, Wc)$
$IV(m)$	$(\alpha, \alpha + \beta, \alpha + 2\beta)$	$m(ab, bc, bW)$
$V(m)$	$(2\alpha, 2\beta, \alpha + \beta)$	$m(aU, bV, c^2)$
$E(m)$	$(\alpha + \beta, \alpha + \beta, \alpha + \beta)$	$m(ab, ab, ab)$

and, with $h = b + c$ and $g = b + 2c$, the ordered Group 1 data are

shape	angles	opposite sides
$S_1(n)$	$(2\alpha, \beta, \alpha + \beta)$	$n(ah, bc, c^2)$
$S_2(n)$	$(2\alpha, \alpha, 2\beta)$	$n(ch, c^2, bg)$
$S_3(n)$	$(\beta, \beta, 3\alpha)$	$n(c^2, c^2, ag)$
$S_4(n)$	$(\alpha + \beta, \alpha + \beta, \alpha)$	$n(bc, bc, ab)$
$S_5(n)$	$(\alpha, \alpha, \alpha + 2\beta)$	$n(bc, bc, bh)$

For Group 2, the attachment scales and count identities in Proposition 17 are:

transfer	patch scale	normalized count identity
$E(m) \to IV(m)$	bm	$ab + b^2 = bW$
$IV(m) \to I(m)$	bm	$bW + b^2 = bU$
$IV(m) \to III(b, a; m)$	Wm	$bW + W^2 = UW$
$V(m) \to II(m)$	Um	$UV + U^2 = 3UW$
$IV(as) \to IV(cs)$	bWs	$bWa^2 + b^2W^2 = bWc^2$
$V(as) \to III(a, b; cs)$	bVs	$UVa^2 + b^2V^2 = VWc^2$
$V(bs) \to III(b, a; cs)$	aUs	$UVb^2 + a^2U^2 = UWc^2$

Here and below the omitted parameter pair is (a, b). The last two identities follow from

$$Ua^2 + b^2V = Wc^2$$

and its $a \leftrightarrow b$ image.

In every case the straight pair is $(\alpha + \beta) + \gamma = \pi$. The shared-side and merged-side checks are:

transfer	shared side	other angle	destination sides
$E(m) \to IV(m)$	$abm = a(bm)$	$(\alpha + \beta) + \beta$	$(abm, bcm, \underbrace{abm + b^2m}_{bWm})$
$IV(m) \to I(m)$	$abm = a(bm)$	$(\alpha + 2\beta) + \beta$	$(bcm, bcm, \underbrace{bWm + b^2m}_{bUm})$
$IV(m) \to III(b, a; m)$	$bWm = b(Wm)$	$\alpha + \alpha$	$(bcm, \underbrace{abm + aWm}_{aUm}, cWm)$
$V(m) \to II(m)$	$aUm = a(Um)$	$2\beta + \beta$	$(c^2m, cUm, \underbrace{bVm + bUm}_{3bWm})$
$IV(as) \to IV(cs)$	$abWs = a(bWs)$	$\alpha + \beta$	$(abcs, \underbrace{a^2bs + b^2Ws}_{bc^2s}, bcWs)$
$V(as) \to III(a, b; cs)$	$abVs = a(bVs)$	$2\alpha + \beta$	$(ac^2s, bcVs, \underbrace{a^2Us + b^2Vs}_{Wc^2s})$
$V(bs) \to III(b, a; cs)$	$abUs = b(aUs)$	$2\beta + \alpha$	$(bc^2s, acUs, \underbrace{b^2Vs + a^2Us}_{Wc^2s})$

For Group 1 put

$$a = pq, \quad b = q^2 - p^2, \quad c = q^2, \quad h = b + c, \quad g = b + 2c.$$

The normalized tables are:

transfer	patch scale	normalized count identity
$S_4(n) \to S_1(n)$	cn	$bc + c^2 = ch$
$S_4(n) \to S_5(n)$	bn	$bc + b^2 = bh$
$S_1(n) \to S_2(n)$	hn	$ch + h^2 = hg$
$S_1(n) \to S_3(n)$	cn	$ch + c^2 = cg$
$S_1(bs) \to S_1(cs)$	ahs	$chb^2 + a^2h^2 = chc^2$
$S_4(ps) \to S_4(qs)$	bqs	$bcp^2 + b^2q^2 = bcq^2$

The corresponding Group 1 side checks are:

transfer	shared side	other angle	destination sides
$S_4(n) \to S_1(n)$	$bcn = b(cn)$	$\alpha + \alpha$	$(\underbrace{abn + acn}_{ahn}, bcn, c^2n)$
$S_4(n) \to S_5(n)$	$abn = a(bn)$	$(\alpha + \beta) + \beta$	$(bcn, bcn, \underbrace{bcn + b^2n}_{bhn})$
$S_1(n) \to S_2(n)$	$ahn = a(hn)$	$\beta + \beta$	$(chn, c^2n, \underbrace{bcn + bhn}_{bgn})$

transfer	shared side	other angle	destination sides
$S_1(n) \to S_3(n)$	$bcn = b(cn)$	$2\alpha + \alpha$	$(c^2n, c^2n, \underbrace{ahn + acn}_{agn})$
$S_1(bs) \to S_1(cs)$	$abhs = b(ahs)$	$\beta + \alpha$	$(achs, bc^2s, \underbrace{b^2cs + a^2hs}_{c^3s})$
$S_4(ps) \to S_4(qs)$	$bcps = a(bqs)$	$\alpha + \beta$	$(bcqs, \underbrace{abps + b^2qs}_{bcqs}, \underbrace{bcps}_{abqs})$

The final identities use

$$b^2c + a^2h = c^3, \qquad abp + b^2q = bcq, \qquad bcp = abq.$$

For completeness, the exhaustion is an angle calculation, not a search over lengths. In Group 2,

$$\pi = 3\alpha + 3\beta, \qquad \gamma = 2\alpha + 2\beta.$$

The complements in π of the three tile angles are respectively $2\alpha + 3\beta$, $3\alpha + 2\beta$, and $\alpha + \beta$. Among target vertices capable of a straightening attachment, only $\alpha + \beta$ occurs, in rows E, IV, and V. Assigning the two remaining tile angles to the two seam endpoints gives exactly the Group 2 rows above, their reflected forms, and one IV-to-reptile output. Similarly, in Group 1 the complements are $2\alpha + 2\beta$, $3\alpha + \beta$, and $\alpha + \beta$. Only $\alpha + \beta$ occurs, in S_1 and S_4; the two assignments give the six listed transfers and the S_1-to-reptile and S_4-to-reptile outputs. The incommensurability hypothesis ensures that no unlisted equality of angle vectors is hidden in these comparisons.

More explicitly, let $\mathcal{O}_j(X)$ be the set of row-shaped angle outputs from a source row X in Group j, let R denote a tile-shaped output, and let * interchange (a, α) with (b, β). The enumeration is

$$\begin{aligned} &\mathcal{O}_2(I) = \mathcal{O}_2(II) = \mathcal{O}_2(III) = \varnothing, \\ &\mathcal{O}_2(E) = \{IV, IV^*\}, \qquad \mathcal{O}_2(IV) = \{I, IV, III^*, R\}, \\ &\mathcal{O}_2(V) = \{II, II^*, III, III^*\}, \end{aligned}$$

and

$$\begin{aligned} &\mathcal{O}_1(S_2) = \mathcal{O}_1(S_3) = \mathcal{O}_1(S_5) = \varnothing, \\ &\mathcal{O}_1(S_1) = \{S_1, S_2, S_3, R\}, \qquad \mathcal{O}_1(S_4) = \{S_1, S_4, S_5, R\}. \end{aligned}$$

For the five non-preserving angle outputs, the generic destination multipliers are

$$\frac{cm}{a}, \quad \frac{cm}{a}, \quad \frac{cm}{b}, \quad \frac{cm}{b}, \quad \frac{qm}{p}.$$

Pairwise coprimality of (a, b, c) and $\gcd(p, q) = 1$ force, respectively, $a \mid m, a \mid m, b \mid m, b \mid m, p \mid m$. These are exactly the as, as, bs, bs, ps parametrizations in Proposition 17; the remaining outputs preserve the normalized multiplier.

C.2. **Orientation-changing parallelograms and the 720 inventory.**

Lemma 24 (Herdt–Beeson parallelogram). *Let $p, q \in \mathbb{N}_0$, not both zero. A parallelogram with adjacent side lengths*

$$bc, \qquad pc + qb$$

and included angle α is tiled by $2(pc + qb)$ copies of R. If $p, q > 0$, its two subparallelograms use different grid orientations. Cyclic side-label variants hold as well.

Proof. Split the second side after length pc. The first subparallelogram is a c-by-p array of fundamental cells with adjacent sides b, c; the second is a b-by-q array with those roles reversed. Each cell splits along its a-diagonal into two copies of R. The shared line may have different edge subdivisions on its two sides, which is permitted because tilings need not be edge-to-edge. □

For $(a, b, c) = (4, 5, 6)$, Herdt's 720-tiling in [3, Fig. 24] has target sides $(120, 120, 180)$ and count $5 \cdot 12^2$. Its vector drawing has ten triangular macroregions: one quadratic patch of scale 15 and three each of scales $6, 5, 4$. These contribute

$$15^2 + 3(6^2 + 5^2 + 4^2) = 456.$$

The inventory has three orientation-changing blocks. Two have common side 30 and width $59 = 24 + 35$, with uniform constituent counts $48 + 70$ in opposite orders; the third is the cyclically relabelled block with common side 20 and width $14 = 4 + 10$, with counts $8 + 20$. Thus the macro count is

$$456 + (48 + 70) + (48 + 70) + (8 + 20) = 720.$$

The rounded source vector paths are consistent with this grouping and total area. Since no exact labelled macro coordinates are archived, this paragraph records an arithmetic audit of a manually transcribed figure inventory rather than a coordinate or coverage certificate.

C.3. **The shape-1/shape-5 complement.** Every shape-4 tiling has $m = qn$ and, after a single scaled tile is attached along its base, produces a shape-5 tiling at multiplier n. Thus shape 4 is subordinate to shape 5. Conversely, a shape-1 and a shape-5 target at the same multiplier glue along a common side to form a reptiling; the remaining difficulty is to force the corresponding seam inside an arbitrary reptiling.

Appendix D. Ordered interior structures

The relation lattice. Let $\rho = e^{i\pi/3}$, so

$$N(x + y\rho) = x^2 + xy + y^2.$$

The standard primitive Eisenstein-triple parametrization gives, after interchanging a, b if necessary,

$$a = r^2 - s^2, \qquad b = 2rs + s^2, \qquad c = r^2 + rs + s^2,$$

where $r > s > 0$, $\gcd(r, s) = 1$, and $r \not\equiv s \pmod 3$. For completeness, the factorization argument is as follows. In the Euclidean domain $\mathbb{Z}[\rho]$,

$$N(a + b\rho) = c^2.$$

The two conjugate factors are coprime up to units: a common nonunit would force a common rational prime divisor of a, b, c, except possibly the ramified prime over 3, and that possibility is excluded by $3 \nmid c$. Unique factorization therefore makes $a + b\rho$ a unit times a square. Multiplication by a unit, conjugation, and if necessary interchange of a, b put its square root in the chamber $r > s > 0$. Expanding $(r + s\rho)^2$ gives the displayed formulae. Common factors of r, s, or $r \equiv s \pmod 3$, would make the resulting triple nonprimitive; the converse is immediate from the same formulae.

Direct expansion now gives

$$rc = (r + s)a + sb, \qquad rb = s(a + c), \qquad (r + 2s)a = (r - s)(b + c).$$

Thus

$$R_c = (r + s, s, -r), \qquad R_b = (s, -r, s)$$

lie in the integer kernel of (a, b, c). Moreover,

$$R_c \times R_b = -(a, b, c).$$

The normal vector on the right is primitive. Hence the parallelogram spanned by R_c, R_b has index one in the rank-two lattice $\ker_{\mathbb{Z}}(a, b, c)$, proving Proposition 18(i). This statement concerns signed, unordered counts of complete a-, b-, and c-edges. It contains no endpoint order, connectivity, or geometric-realizability assertion.

Pure-edge support words. For an oriented a-edge $[x, x + a]$ supporting a tile in the upper half-plane, the two configurations in Proposition 18 have endpoint labels

$$A : (\beta, \gamma), \qquad B : (\gamma, \beta).$$

An $A \to B$ junction would put two $\gamma = 2\pi/3$ sectors in the same half-plane and is impossible. Therefore a support word contains no substring AB, which is equivalent to the form $B^j A^{n-j}$.

At an $A \to A$ or $B \to B$ junction, the supported angles sum to $\beta + \gamma = \pi - \alpha$, leaving an α sector. If that sector contains $n_\alpha, n_\beta, n_\gamma$ tile angles, then

$$(n_\alpha - n_\beta - 1)\alpha + \frac{n_\beta + 2n_\gamma}{3}\pi = 0,$$

because $\beta = \pi/3 - \alpha$ and $\gamma = 2\pi/3$. The irrationality of α/π, proved in the next paragraph, gives $n_\alpha = 1$ and $n_\beta = n_\gamma = 0$. This forces one tile angle, but it does not decide which of that tile's two adjacent sides lies on which wall; that distinction is the forced-layer mirror gap.

At the possible $B \to A$ junction, put the common endpoint at $O = 0$. The two supported third vertices are

$$Q_- = -a + b\rho^2, \qquad Q_+ = a + b\rho.$$

Using $\rho^2 = \rho - 1$ gives

$$|OQ_-| = |OQ_+| = c, \qquad |Q_-Q_+| = 2a + b.$$

Both third vertices have height $b\sqrt{3}/2$, so the defect area is $(2a + b)b\sqrt{3}/4$, whereas the tile area is $ab\sqrt{3}/4$. Their ratio is $(2a + b)/a$. If this were integral, then $a \mid b$; primitivity would force $a = 1$. But then

$$(2c - (2b + 1))(2c + (2b + 1)) = 3$$

forces $b = 0$, contrary to positivity. The defect wedge is therefore never a union of whole tiles.

The directed c-flow. Orient every c-edge from its α endpoint to its β endpoint, splitting a through-edge at each T-junction. At a non-corner vertex, let A, B, G count incident α, β, γ tile corners, and let F count incident tiles for which the vertex lies in the relative interior of a tile edge. Each such tile contributes a flat sector of angle π. Since $\gamma = 2(\alpha+\beta)$ and $\pi = 3(\alpha+\beta)$, the angle equation is

$$(A+2G)\alpha + (B+2G)\beta = (6-3F)(\alpha+\beta) \quad \text{or} \quad (3-3F)(\alpha+\beta),$$

according as the vertex is interior or lies straight on the target boundary. Here α/π is irrational. Indeed, $2\cos\alpha = U/c$ is rational and lies strictly between 1 and 2, because $U^2 - c^2 = 3bW > 0$ and $4c^2 - U^2 = 3a^2 > 0$. If α/π were rational, $2\cos\alpha$ would be a rational algebraic integer and hence an integer, a contradiction. Hence α and $\beta = \pi/3 - \alpha$ are linearly independent over $\mathbb{Q}$, and $A = B$. An α corner contributes one outgoing c-end, a β corner one incoming end, and a γ corner neither. A through-c edge contributes one incoming and one outgoing end, while a through-a or through-b edge contributes neither. The flow is therefore conserved. At a convex target corner no flat sector can occur. If its angle is $P\alpha + Q\beta$, then

$$A + 2G = P, \qquad B + 2G = Q,$$

so its net outflow is $A - B = P - Q$. Standard decomposition of the resulting finite directed multigraph gives directed cycles and source-to-sink c-edge paths. The latter may run along the target boundary; no interiority claim is made.

The row-III half-parallelogram. Let $\rho = e^{i\pi/3}$, $u = b$, and $v = b + a\rho$; then $|v| = c$ and the (u, v) unit parallelogram is two copies of the tile because $v - u = a\rho$. The multiplier-one row-III target is

$$T = \operatorname{conv}(0, Vu, Wv).$$

Indeed, its first two side lengths are bV and cW, while

$$Wv - Vu = a(W\rho - b), \qquad |W\rho - b|^2 = W^2 - Wb + b^2 = c^2,$$

so its third side is ac. Its area is VW tile areas. Thus T is one half of the V-by-W parallelogram tiled in the (u, v) grid. Moreover the cut from Vu to Wv is a straight chain of a legal c-edges in the second orientation. What remains for a uniform construction is precisely to fill across this change of grids; the exact 88-tiling does so for $(a, b, c) = (3, 5, 7)$.

Appendix E. Result ledger and computational qualifications

Table 4 summarizes the results most relevant to a final solution. It does not claim that the displayed values form a complete frontier.

status	values or families	basis
members	the five classical families; published sporadic constructions; newly $88m^2$ and $189m^2$	classical theorems, cited constructions, exact certificates
nonmembers	$7, 11, 14, 15$; every prime $p > 3$ with $p \equiv 3 \pmod 4$; and $21, 30, 33, 46$	published proofs, proved arithmetic, or scoped exact searches
further audited exclusions	$55, 56, 66, 70, 438$	exact arithmetic audits; 56 and 70 include single-implementation exhaustive searches
finite-window ray problems	every Group 2 ray	Corollary 12
additive ray problems	Group 1 shapes 4–5; no cofinite tail is proved in general	Proposition 14
Group 2 rays with unresolved finite windows	for example the rays with base counts 33, 154, and 184	only finitely many multipliers remain undecided on each; the χ-based ideal-membership arithmetic is already exhausted

Table 4. Conservative status summary.

The table below intentionally omits the forced-layer-dependent exclusions. It also distinguishes "separately derived" from merely running two programs in the same repository: the 88 and 189 certificates have a checker independent of the search history, but a third-party reimplementation remains desirable.

The phrase "full branch audit" means that the classical, right-triangle, Group 1, Group 2, $\gamma = 2\alpha$, and $60°$ equilateral branches were all enumerated under the cited classification, rather than that one displayed congruence was treated as a global exclusion. The executable audits are in `computation/verify_theory_report3_claims.py` (including the explicit $N = 33$ reduction and the audits of $21, 30, 55, 66, 438$) and `computation/sieve_120rows.py`; their commands and hashes are in `paper/verification_manifest.md`.

N	present basis for nonmembership	evidence
7, 11	Beeson's direct proofs [1]	published
14, 15	Beeson's finite equilateral analysis [1]; 14 also has a project exact exhaustion	published + search
$p \equiv 3 \pmod 4$	three prime-case manuscripts [10, 9, 5]	independently derived
21	full branch audit; exact exhaustion of the surviving Group 1 shape-5 candidate	arithmetic + search
30	full branch audit; the Group 2 candidate fails boundary divisibility, with an exact exhaustion as a cross-check	arithmetic + search
33	unique candidate; primary exact exhaustion, with a replay disabling the general segment-semigroup table and boundary-invariant pruning	computer-assisted, one implementation
46	complete arithmetic branch audit; also [9]	independently derived
55	full branch audit; the Group 2 row-IV candidate has $b \nmid k$	proved + machine-audited arithmetic
56	full candidate audit; row-E and two shape-4 exact exhaustions	computer-assisted, decisive searches not independently ported
66	full branch audit; the Group 2 row-E candidate has $ab \nmid L$	proved + machine-audited arithmetic
70	full candidate audit; shape-5 exact exhaustion	computer-assisted, decisive search not independently ported
438	full arithmetic branch audit using the audited Group 1 shape-5 formula	machine-audited arithmetic

Appendix F. Provisional below-200 research programme

Not a theorem. Everything in this appendix depends on an unfinished forced-corner-layer induction and on a complete regeneration of the cross-branch candidate list. It is reported to direct future work and must not be cited as an established reduction.

The proposed forced-layer lemma would make a distinguished corner patch repeat along the equal sides of certain isosceles targets. The supporting line, chord partition, purity, and arithmetic residual steps have been checked. The missing step must exclude the mirror side-assignment of the tile placed in each interstitial gap and justify the endpoint-sector order. An undershooting mirror placement leaves angle-legal completions; an overshooting placement crosses the next chord line. No completed argument currently rules out every intermediate configuration.

A subsequent attempted proof correctly partitions the existing chord into coherently oriented tile edges and identifies the ideal triangular regions between two consecutive scaled patches. It then infers that the actual tiling occupies those ideal regions. That inference is precisely the missing step: the reflected congruent placement exchanges the two adjacent side lengths, undershoots one ideal wall at a T-junction, and crosses the artificial next chord on the other side. The pure-edge word theorem in Proposition 18 forces the residual angle at a coherent junction, but not this side assignment. Thus the later attempt does not complete the forced-layer induction or establish any of its consequences.

If the forced-layer induction and full candidate regeneration are completed as intended, the provisional ledger predicts that the unresolved values below 200 reduce to

$$105,\ 120,\ 132,\ 143,\ 154,\ 161,\ 175,\ 184.$$

At present, the status of at least

$$\{60, 63, 76, 92, 99, 105, 120, 124, 132, 135, 140, 143, 154, 156,\\ 161, 171, 172, 175, 184, 188\}$$

is materially entangled with the missing induction or the audit built on it. In particular, neither "only eight values remain below 200" nor "every $N \le 104$ is decided" is asserted here.

F.1. **Provisional candidate table.** Assuming the forced-layer induction and complete below-200 candidate regeneration, the provisional eight-value table is:

N	proposed surviving instance(s)
105	row IV: $(8, 7, 13)$ at $k = 7$; $(16, 5, 19)$ at $k = 5$
120	row IV: $(7, 8, 13)$ at $k = 8$; a conditional $60°$ skeleton also remains
132	row I: $(5, 3, 7)$ at multiplier 2
143	row V: $(3, 5, 7)$ at multiplier 1
154	row I: $(8, 7, 13)$ at multiplier 1
161	Group 1 shape 5: $(12, 7, 16)$, $(p, q) = (3, 4)$, multiplier 1
175	$\gamma = 2\alpha$: $(9, 7, 12)$, $(u, v) = (3, 4)$, $r = 5$
184	row I: $(7, 8, 13)$ at multiplier 1

Positive transfer tests can be cheaper than direct searches: a row-IV $(5, 3, 7)$ tiling at $N = 96$ would imply $132 \in \mathcal{S}$; the remaining Group 1 shape-4 instance at $N = 112$ would imply $161 \in \mathcal{S}$; and row-IV finds at 105 or 120 would also settle the linked values 154 or 184, respectively. These are one-way construction opportunities, not nonexistence proofs.

F.2. **Priorities for deconditioning the ledger.** The most direct next steps are:

(1) close the forced-layer mirror gap with a side-assignment argument valid in the presence of T-junctions, together with the endpoint-sector order;
(2) independently exhaust the $\gamma = 2\alpha$ instances at $N = 63$ and $N = 99$, and the $60°$ candidates at $N = 105$;
(3) port the exact searcher to the $\gamma = 2\alpha$ geometry and attempt the sole proposed $N = 175$ instance; and
(4) produce a script-generated all-branch candidate ledger through 200, with every imported theorem and conditional filter separately switchable.

Disclaimer: use of AI tools

This project used multiple AI systems extensively. OpenAI GPT-5.6 Pro agents and Anthropic Claude Fable agents proposed lemmas, derived formulae, wrote and audited exact-search code, compared source versions, and helped draft this manuscript. Separate agents were assigned mathematical derivation, computation, source checking, and adversarial review.

Agreement among models is an error filter, not independent scholarly verification. Every unconditional theorem is either proved in the text, imported with a precise citation, or explicitly labelled computer-assisted with its verification surface stated. Responsibility for any circulated version remains with the author.

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